

# Jingeun Kim

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## RESEARCH INTERESTS

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My research primarily focuses on developing efficient machine learning frameworks. Specifically, I aim to explore efficient models using neural architecture search based on genetic algorithms. Furthermore, I am currently working on model compression, particularly pruning, using genetic algorithms for CNN-based models and pre-trained models.

**Keywords:** Neural Architecture Search, Genetic Algorithm, Model Compression

## EDUCATION

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Doctor of Philosophy:

Gachon University, Gyeonggi-do, South Korea (Mar. 2024 - )

Major in IT Convergence Engineering (Advisor: Yourim Yoon)

4.39/4.50 GPA

Master of Science:

Gachon University, Gyeonggi-do, South Korea (Sep. 2021 - Aug. 2023)

Major in IT Convergence Engineering (Advisor: Yourim Yoon)

4.06/4.50 GPA

Thesis title: Machine Learning with Various Feature Selection Methods for Diagnosis of Parkinson's Disease Using RNA-Seq

Bachelor of Science:

Gachon University, Gyeonggi-do, South Korea (Mar. 2016 - Aug. 2021)

Major in Computer Engineering (Advisor: Yourim Yoon)

3.73/4.50 GPA

## WORK EXPERIENCES

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Gachon University

Sep 2023 – Feb 2024

Master's Degree Researcher

- Optimization of DenseNet architecture using a genetic algorithm with multi-dimensional encoding scheme

## PROJECTS

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Project 1 (May 2020 - May 2021)

**Analysis of Pulmonary Diseases Induced by Fine Particulate Matter and Treatment Drug Screening Using Meta-Heuristic-Based Feature Selection Algorithm**

Summary: This study compares how machine learning is efficient when classifying mRNAs or miRNAs.

Role: model implementation, evaluation, interpretation, and manuscript writing

Advisor: Yourim Yoon

Status: Paper accepted

Project 2 (Mar. 2021 - Nov. 2021)

**Development of Next-Generation Battery Management System (BMS) with AI-Based Fire and Explosion Prevention for Lithium-ion Batteries**

Summary: Estimate the resistance value of the battery cell after 10 days to prevent explosion and Battery anomaly detection using clustering.

Role: Researcher (model implementation, evaluation)

Advisor: Taegkeun Whangbo

Status: Paper accepted

Project 3 (May 2021 - May 2022)

**Theory and Applications of Various Feature Selection Algorithms**

Summary: This study uses EEG datasets and genetic algorithms to select meaningful features to increase the accuracy of machine learning.

Role: Overall study design, model implementation, evaluation, interpretation, and manuscript writing

Advisor: Yourim Yoon

Status: Manuscript submitted

Project 4 (May 2022 – July 2024)

**Research on IT-BT/ET Convergence Technology Using Meta-Heuristic-Based Feature Selection Algorithms**

Summary: Various feature selection methods improve the performance of machine learning when classifying Parkinson's disease

Role: Overall study design, model implementation, evaluation, interpretation, and manuscript writing

Advisor: Yourim Yoon

Status: Paper accepted

Project 5 (May 2023 – July 2025)

**Healthway Development for Senior Citizens in the Age of Homo sapiens [Re:]Solution: Building a Healthway for Muscle Atrophy Prevention and Management**

Summary: Enhancing classification accuracy through feature selection using correlation analysis

Role: Overall study design, model implementation, evaluation, interpretation, and manuscript writing

Advisor: Yourim Yoon

Status: Manuscript submitted

Project 6 (Jan 2024 - Present)

**AI-based Maritime Search and Rescue Decision Making**

Summary: This study utilizes AI to improve real-time data analysis, predictive modeling, and resource optimization, enhancing the efficiency and effectiveness of maritime SAR operations.

Role: Overall study design, model implementation, evaluation, interpretation, and manuscript writing

Advisor: Yourim Yoon

Status: Work in progress

Project 6 (Sep 2025 - Present)

**Evolutionary Algorithm-Based Lightweight Techniques for Large Language Models**

Summary: This project investigates evolutionary algorithm-driven structural optimization

methods for large language models (LLMs) to achieve lightweight architectures while preserving inference accuracy

Role: Principal investigator, Overall study design, experimental evaluation, performance analysis, and manuscript preparation.

Funding: 50 million KRW ( $\approx$  USD 38,500) over two years, awarded through the Ph.D. Student Scholarship Program.

Status: Work in progress

## PUBLICATIONS

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- J6. Pruning for Efficient DenseNet via Surrogate-Model-Assisted Genetic Algorithm considering NAS Proxies  
**Jingeun Kim**, Yourim Yoon  
Swarm and Evolutionary Computation
- J5. A Comparison of Binary and Integer Encodings in Genetic Algorithms for the Maximum k-Coverage Problem with Various Genetic Operators  
Yoon Choi, **Jingeun Kim**, Yourim Yoon  
Biomimetics
- J4. Performance Analysis and Improvement of Machine Learning with Various Feature Selection Methods for EEG-Based Emotion Classification  
Sherzod Abdumalikov, **Jingeun Kim**, Yourim Yoon  
Applied Sciences
- J3. Improving Modularity Score of Community Detection Using Memetic Algorithms  
Dongwon lee, **Jingeun Kim**, Yourim Yoon  
AIMS mathematics
- J2. RNA Sequences-Based Diagnosis of Parkinson's Disease Using Various Feature Selection Methods and Machine Learning  
**Jingeun Kim**, Hyejin Park, Yourim Yoon  
Applied Sciences
- J1. Comparative Study of Classification Algorithms for Various DNA Microarray Data  
**Jingeun Kim**, Yourim Yoon, Hyejin Park, Yong-hyuk Kim  
Genes

## CONFERNCES

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- C4 Sparse Transformer for Long-term Forecasting via Genetic Algorithm  
**Jingeun Kim**, Yourim Yoon  
GECCO 2025 (poster)
- C3. Efficient Pruning of DenseNet via a Surrogate-Model-Assisted Genetic Algorithm  
**Jingeun Kim**, Yourim Yoon  
GECCO 2024 (poster)
- C2. Genetic Algorithm-based Pruning for Efficient DenseNet Architecture  
**Jingeun Kim**, Yong-hyuk Kim, Yourim Yoon  
ICAIIC 2024
- C1. the battery risk prediction method using the deep learning  
Kyunam Choi, Mona O, **Jingeun Kim**, Taegkeun Whangbo  
ITC-CSCC 2021

## WORKSHOPS

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- W1. From Forecast to Action: Uncertainty-Aware UAV Deployment for Ocean Drifter Recovery  
**Jingeun Kim**, Yong-Hyuk Kim, Yourim Yoon  
CIKM Workshop on Spatio-Temporal Data Intelligence and Foundation Models 2025

## DOMESTIC PUBLICATIONS

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- J2. Enhancing Machine Learning-based Drug-Drug Interaction Prediction using Feature Selection  
Hojae Kim, **Jingeun Kim**, Yourim Yoon  
Asia-pacific Journal of Convergent Research Interchange
- J1. Battery thermal runaway cell detection using DBSCAN and statistical validation algorithms  
**Jingeun Kim**, Yourim Yoon  
The Journal of the Convergence on Culture Technology

## MANUSCRIPTS UNDER REVIEW AND IN PREP

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- C1. MAC: Memetic Actor-Critic Framework for UAV Path Planning  
**Jingeun Kim**, Seung-ryul Hong, Yong-Hyuk Kim, Yourim Yoon  
GECCO 2026
- J1. A Study on the Effectiveness of Feature Selection considering mRMR across Different Datasets  
**Jingeun Kim**, Yourim Yoon  
Knowledge and Information Systems

## TEACHING EXPERIENCE

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|                    |                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------|
| <b>2025 Fall</b>   | JAVA @Gachone University (T.A.)                                                                  |
| <b>2025 Spring</b> | Algorithm @Gachone University (T.A.)                                                             |
| <b>2024 Fall</b>   | Discrete mathematics @Gachone University (T.A.)                                                  |
| <b>2024 Spring</b> | Algorithm @Gachon University (T.A.)                                                              |
| <b>2023 Fall</b>   | Discrete mathematics @Gachon University (T.A.)<br>Advanced Programming @Gachon University (T.A.) |
| <b>2023 Spring</b> | Algorithm @Gachon University (T.A.)                                                              |
| <b>2022 Fall</b>   | Discrete mathematics @Gachon University (T.A.)                                                   |
| <b>2022 Spring</b> | Algorithm @Gachon University (T.A.)                                                              |

## AWARDS, HONORS & SCHOLARSHIPS

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|-----------|---------------------------------------------------------------------------------|----------|
| 2025-2027 | Ph.D. Student Scholarship Program                                               | 38,500\$ |
| 2025      | SCI best paper award                                                            | 350\$    |
| 2024-     | Tuition Exemption (100%), Educational Funding Scholarship                       | 4500\$   |
| 2021-2023 | Tuition Exemption (100%), Educational Funding Scholarship<br>@Gachon University | 4500\$   |
| 2022      | SCI best paper award<br>@Gachon University                                      | 100\$    |
| 2020      | Gachon University x-design contest 3 <sup>rd</sup> place<br>@Gachon University  | 200\$    |
| 2016-2021 | Tuition Exemption (30%), Educational Funding Scholarship                        | 1000\$   |

@Gachon University

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**PROFESSIONAL SERVICE**

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|------------------|----------------------------------------------------------------------------------|
| Journal Reviewer | IEEE Transactions on Evolutionary Computation, BMC Psychiatry, Discover Oncology |
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**SKILLS**

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| Programming Languages | Python, JAVA                                |
| AI Frameworks         | Pytorch, Tensorflow, sklearn, pandas, numpy |
| Languages             | Korean (Native), English (Intermediate)     |

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**REFERENCES**

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|---------|-----------------------------------------------------------------------------------|
| Advisor | Prof. Yourim Yoon<br><a href="mailto:yryoon@gachon.ac.kr">yryoon@gachon.ac.kr</a> |
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